



GANPAT UNIVERSITY

FACULTY OF COMPUTER APPLICATIONS

Programme	Master of Computer Applications					Branch/Spec.	Computer Application		
Semester	II					Version	1.0.0.0		
Effective from Academic Year			2022-23			Effective for the batch Admitted in		June 2022	
Subject Code	P12A5AIR		Subject Name			Artificial Intelligence and Robotics			
Teaching scheme						Examination scheme (Marks)			
(Per week)	Lecture (DT)		Practical (Lab.)		Total		CE	SEE	Total
	L	TU	P	TW					
Credit	2	0	2	0	4	Theory	40	60	100
Hours	2	0	4	0	6	Practical	20	30	50

Objective:

- The objective of the course is to basic concepts of artificial intelligence (AI) principles and approaches.
- Develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents: Search, Knowledge representation, inference, logic, and learning.
- To describe the strengths and limitations of various state-space search algorithms, and choose the appropriate algorithm.
- To understand the basic concepts associated with the design and functioning and applications of Robots To study about the drives and sensors used in Robots

Pre-requisites:

- Data Structure, Basic probability theory and Statistics, Knowledge of programming language.

Course Outcomes :

- 1 = Slight (Low); 2 = Moderate (Medium); 3 = Substantial (High); “-” = No Correlation

Name of CO	Description											
CO1	Understand the theory of Artificial intelligence, Intelligent Agents											
CO2	Learn Prolog Programming language and implement AI algorithm.											
CO3	Understand Uninformed Search Strategies and apply BFS , DFS and Informed Search Strategies Heuristic Functions, Best-First Search, Greedy Search, A* Algorithm,											
CO4	To learn about application of Robots, knowledge for the design of robotics, robot kinematics and robot programming, Robotics Actuators, Sensors and Robot Control											
	Mapping of CO and PO											
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	3	3	2	2	2	2	1	1	1
CO2	3	3	3	3	2	2	2	2	2	1	2	-

Q.1 –8 Marks

Q.2 –11 Marks

Q.3 –11 Marks

SECTION - II

Q.4 –8 Marks

Q.5 –11 Marks

Q.6 –11 Marks